

Vincent Yu, Ph.D.

Functional Genomics Leadership | Drug Discovery | AI-Enabled Biological Discovery

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Summary of Qualifications

Mechanistic biologist with deep experience spanning pulmonary disease biology, sensory neuroscience, and oncology, complemented by expertise in functional genomics, CRISPR screening, genome editing, NGS, and single-cell/multi-omic analysis. Experienced in using experimental and computational approaches to uncover disease mechanisms, identify and validate therapeutic targets, and translate complex biological data into actionable program hypotheses. Proficient in Python/R and experienced in using AI-enabled scientific workflows and agentic systems as tools for biological reasoning, evidence integration, and target discovery. Brings a cross-disciplinary perspective connecting molecular and cellular biology, genomics, computation, and drug discovery.

Research Experience

Systems Builder

First-in-Class.ai
Aug 2026 – present

First-in-Class.ai is an independent garage R&D project extending my experience in mechanistic biology, functional genomics, and drug discovery through AI-enabled scientific tools.

- Built an end-to-end [biomedical literature due-diligence workflow](#) integrating claim decomposition, PubMed/full-text retrieval, hybrid search, evidence verification, and grounded synthesis.
- Developing a [public IND-Demystified resource](#) to map late discovery, IND-enabling studies, and first-in-human entry across DMPK, toxicology, CMC, biomarker strategy, and regulatory interactions.
- Exploring real-world use cases through conversations and small prototypes with scientists and biotech professionals.

Visiting Scientist

Trey Ideker Lab
Feb 2026 – present

Applying AI, machine learning, and functional genomics to build computational systems that improve biological reasoning, integrate heterogeneous biomedical data, and accelerate mechanistic discovery.

- Develop, evaluate, and improve agentic systems for biomedical research in collaboration with the Ideker Lab's software engineering team.
- Conduct research at the intersection of functional genomics, systems biology, and AI-enabled biomedical discovery. Integrate computational modeling, CRISPR perturbation screens, and multimodal data to investigate disease mechanisms and therapeutic response.

Senior Scientist Team Lead

Light Horse Therapeutics
San Diego, CA
Jan 2025 – Jan 2026

Built and led functional genomics platforms to accelerate target discovery and precision drug development, integrating technology development, scientific leadership, and cross-functional program execution.

- Built and led a functional genomics platform spanning high-throughput CRISPR screening, base-editor tiling, NGS workflows, and custom computational analysis, establishing core infrastructure for target discovery and mechanism-

Scientist II*Light Horse**Therapeutics**San Diego, CA**Apr 2023 – Jan 2025*

focused drug programs.

- Led a genome-wide CRISPR knockout screening program end-to-end, owning scientific strategy, experimental design, execution, resourcing, and cross-functional integration across biology, sequencing, and computational analysis.
- Identified and validated functional sites on high-priority protein targets, helping advance two preclinical discovery programs.
- Generated core experimental data supporting a multi-year strategic partnership with Novartis, linking functional genomics outputs to external partnership value.
- Partnered with Discovery Biology, Chemistry, proteomics, and Bioinformatics to integrate functional genomics into target evaluation, mechanism-of-action studies, and program strategy.
- Designed and interpreted multi-omics studies—including CUT&RUN, ChIP-seq, ATAC-seq, RNA-seq, and single-cell sequencing—and developed custom Python workflows for CRISPR editing and screening analysis.

Scientist I*Fate Therapeutics**San Diego, CA**Apr 2022 – Apr 2023***Integrated experimental genomics and computational analysis to evaluate genome-editing specificity and support the development of therapeutic cell engineering platforms.**

- Led experimental and computational assessment of CRISPR off-target editing in allogeneic iPSC programs, integrating SITE-seq, GUIDE-seq, computational prediction, and multiplexed amplicon sequencing.
- Adapted SITE-seq for Cas12a-family nucleases, including Cpf1 and Mad7, and developed a computational workflow to quantify editing events from NGS data.
- Developed customized whole-genome sequencing pipelines for variant detection in edited iPSC clones and matched donors using Mutect2 and Strelka2.

Postdoctoral Fellow*UCSD, PI: Xin Sun**San Diego, CA**Aug 2020 – Apr 2022***Applied single-cell genomics, computational biology, and imaging technologies to investigate tissue regeneration and human disease.**

- Investigated lung regeneration following pneumonectomy using single-cell RNA-seq, single-cell ATAC-seq, confocal imaging, and lineage tracing, identifying a previously unrecognized transient mesenchymal cell state associated with alveolar regeneration.
- Led computational analysis of human single-nucleus RNA-seq and ATAC-seq datasets for a multi-institutional consortium studying bronchopulmonary dysplasia.
- Collaborated with neuroscientists to investigate vagal sensory neurons and brainstem circuits regulating allergen-induced airway responses, integrating molecular, imaging, and genomic approaches.

Graduate Research Assistant

PI: Neil Segil

Aug 2014 – Aug 2020

Developed and applied low-input genomic technologies to enable mechanistic studies of gene regulation and biological discovery.

- Optimized low-input ATAC-seq and CUT&RUN protocols for profiling chromatin accessibility and transcription-factor binding in FACS-purified primary cells, enabling epigenomic analysis from limited biological samples.
- Discovered a feed-forward epigenetic mechanism in which transcription factor ATOH1 and pioneer factor POU4F3 cooperate to drive mechanosensory cell differentiation.

Scientific Toolkit

Experimental Biology & Genomics

CRISPR/Cas gene editing and tiling screens, base editing, shRNA, siRNA, mouse breeding/genotyping, PCR, qPCR, ddPCR, cell culture, molecular cloning, lentivirus and AAV production, RNA-seq, ATAC-seq, ChIP-seq, CUT&RUN, immunohistochemistry, western blot, protein purification, FACS, 10x single-cell.

Computational Biology and AI

Linux; Python; R; AWS; NGS analyses for CRISPR editing and screening, RNA-seq, ATAC-seq, ChIP-seq, CUT&RUN, single-cell sequencing, and custom DNA sequencing; agentic workflows, RAG, literature retrieval and LLM-enabled scientific workflows, machine learning for biological datasets

Education

Ph.D., Hearing Neuroscience Program, University of Southern California, 2020
M.S., Cell Biology, Ocean University of China, 2014
B.S., Biology, Ocean University of China, 2011

Publications

1. **Yu HV***, Tao L*, Llamas J, Wang X, Nguyen JD, Segil N. POU4F3 pioneer factor activity enables ATOH1 to drive diverse mechanoreceptor differentiation through a feed-forward epigenetic mechanism. **PNAS** 2021.
2. Tao L*, **Yu HV***, Llamas J, Wang X, Stojanova Z, Segil N. Enhancer decommissioning during inner ear maturation imposes an epigenetic barrier to sensory hair cell regeneration. **Dev Cell** 2021.
3. Xu P*, **Yu HV***, Tseng KC, Flath M, Fabian P, Segil N, Crump JG. Foxc1 established enhancer accessibility for craniofacial cartilage differentiation. **eLife** 2021.
4. Su Y, Xu J, Zhu Z, Chin J, Xu L, **Yu HV** et al. Brainstem Dbh+ neurons control allergen-induced airway hypersensitivity. **Nature** 2024.